```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

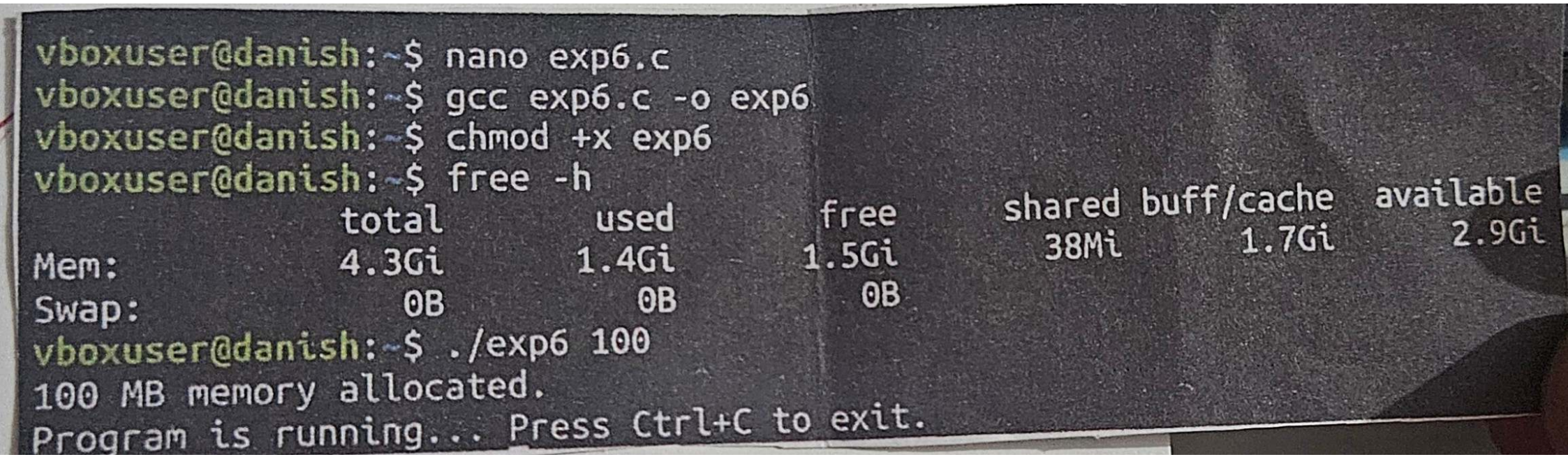
int main(int argc, char *argv[])
{
int mb;
if(argc != 2)
{
printf("Usage: %s <memory_in_MB>\n", argv[0]);
return 1;
}
mb = atoi(argv[1]);

char *memory = malloc(mb * 1024 * 1024);

if (memory == NULL)
{
printf("Memory allocation failed\n");
return 1;
}
for(int i = 0; i < mb * 1024 * 1024; i += 4096)
{
memory [i] = 1;
}
printf("%d MB memory allocated.\n", mb);
printf("Program is running... Press Ctrl+C to exit.\n"

while (1)
{
sleep(1);
}
return 0;
}
```


vboxuser@danish:~$ free -h
total used free shared buff/cache availabi
Mem: 4.3Gi 1.5Gi 1.4Gi 38Mi 1.7Gi 2.8
Swap: 0B 0B 0B
vboxuser@danish:~$


vboxuser@danish:~$ nano exp6.c
vboxuser@danish:~$ gcc exp6.c -o exp6
vboxuser@danish:~$ chmod +x exp6
vboxuser@danish:~$ free -h
total used free shared buff/cache available
Mem: 4.3Gi 1.4Gi 1.5Gi 38Mi 1.7Gi 2.9Gi
Swap: 0B 0B 0B
vboxuser@danish:~$ ./exp6 100
100 MB memory allocated.
Program is running... Press Ctrl+C to exit.